

Program 5

Demonstrate CI Integration by Running Jenkins in a Docker Container

Program Structure:

program5

----->Dockerfile

----->docker-compose.yml

Step 1: Create the a Dockerfile without any extensions and add the following the content
nano Dockerfile

```
GNU nano 6.2 Dockerfile
#use jenkins base
FROM jenkins/jenkins:lts

# Switch to root user to install tools
USER root

#Install Docker inside jenkins container
RUN apt-get update && apt-get install -y docker.io

# Switch back to jenkins user
USER jenkins
```

Step 2 : Create a docker-compose.yml file with following code
nano docker-compose.yml

```
GNU nano 6.2 docker-compose.yml
version: '3.8'

services:
  jenkins:
    build:
      context: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
```

Step 3 : Build an image

docker build -t jenkins .

Run the container

docker run -p 8080:8080 jenkins

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/jenkins$ docker build -t jenkins .
[+] Building 615.3s (6/6) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 274B                                0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts    4.5s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io    0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af0 128.9s
=> => resolve docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004 0.0s
=> => sha256:68cdef48009293f6a12f5c50b91fcfe4194bd6293334d8a31a45b1ed63cb91a5 1.23kB / 1.23kB 0.3s
=> => sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3 65.08MB / 65.08MB 68.6s
=> => sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 5.24MB / 5.24MB 3.8s
=> => sha256:b768bbb699b5f1b6579e23e7253ae1157a339a20a2e9befdf8e03923e893b800 2.58kB / 2.58kB 0.0s
=> => sha256:75109de20007f9345278f298f4534d443409e234ba81225815caa812cc60564 12.71kB / 12.71kB 0.0s
=> => sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 3.12kB / 3.12kB 0.0s
=> => sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168 183B / 183B 1.3s
=> => sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d80688c35e7f1ed86 95.32MB / 95.32MB 91.3s
=> => sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaac0b98 189B / 189B 4.1s
=> => sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fdf88e9c009f6874e7221acb06245 6.29MB / 6.29MB 82.4s
=> => sha256:d7dd2e63023897146ec4b913a460382968fbdd1cc3682e36f650727a1bfd6 72.05MB / 72.05MB 126.9s
=> => extracting sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc 5.2s
=> => extracting sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 0.2s
=> => extracting sha256:68cdef48009293f6a12f5c50b91fcfe4194bd6293334d8a31a45b1ed63cb91a5 0.0s
=> => extracting sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168 0.0s
```

step 4: verify localhost details on browser

<http://localhost:8080>

